BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Biology (Objective Type) — Inter-(Part-II)-A-2021 (For all sessions) — Paper Code: 8465

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8465

Note: NOTE: Write answers to the questions on the objective answer sheet provided. Four possible answers A,B,C and D to each question are given. Which answer you consider correct, fill the corresponding circle A,B,C or D given in front of each question with Marker or pen ink on the answer sheet provided.

Q1. The division of nucleus during cell division is called:

(A) cytokinesis
(B) Karyokinesis
(C) Parthenogenesis
(D) Karyotype

Q2. The crossing over occur in_____stage:

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diplotene

Q3. A gamete without any sex chromosome is:

(A) Heterogamete
(B) Homogamete
(C) Nullogamete
(D) Isogamete

Q4. The plasmid psc_{101} has antibiotic resistance gene for:

(A) Tetracycline
(B) Ampicillin
(C) Penicillin
(D) Terramycin

Q5. Archaeobacteria can tolerate temperature upto:

(A) $118^{o}C$
(B) $119^{o}C$
(C) $120^{o}C$
(D) $121^{o}C$

Q6. The organism, which inhibit the root nodules of legume plants are:

(A) Fungi
(B) Algae
(C) Bacteria
(D) Cynobacteria [sic]

Q7. The grass land in tropical climate having woody trees are called:

(A) Prairies
(B) Savanna
(C) Tundra
(D) Alpine

Q8. Establiishment [sic] of new forests where no forest existed is known as:

(A) Afforestation
(B) Reforestation
(C) Forestation
(D) Deforestation

Q9. The active up take of sodium in ascending limb of loop of Henle is promoted by_____hormone:

(A) Aldosterone
(B) ADH
(C) Testosterone
(D) Progesterone

Q10. Which one of the following is an ectotherm:

(A) Bird
(B) Huming [sic] bird
(C) Amphibian
(D) Bat

Q11. The active conducting portion of wood in older trees is:

(A) Sap wood
(B) Heart wood
(C) Bark
(D) Callus

Q12. Arthritis is an inflammatory or degenerative disease that damage:

(A) Muscles
(B) Brain
(C) Joints
(D) Kidney

Q13. The part of brain, which play role in the formation of long term memory is:

(A) Thalamus
(B) Hippocampus
(C) Amygdala
(D) Pons

Q14. Fruit development without fertilization is called:

(A) Vernalization
(B) Parthenogenesis
(C) Parthenocarpy
(D) Dormancy

Q15. Which colour cytoplasm of an ascidian fertilized egg gives rise gut_____:

(A) Clear cytoplasm
(B) Yellow cytoplasm
(C) Grey equatorial cytoplasm
(D) Grey vegetal cytoplasm

Q16. The ability to regain the lost or injured part of the body is called:

(A) Aging
(B) Regeneration
(C) Generation
(D) Degeneration

Q17. Which of the following is initiation codon?

(A) AUG
(B) UAA
(C) UGG
(D) UGA

Biology (Essay Type) — Inter-(Part-II)-A-2021 (For all sessions)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Q. Section-I

Q2. Write short answers of any eight parts from the following.

2x8=16

(i) What is peritonial dialysis?

(ii) What is panting?

(iii) Differentiate between Poikilotherms and Homeotherms.

(iv) What is Ecdysis?

(v) Differentiate between Hyaline cartilage and Elastic cartilage.

(vi) What is Sciatica?

(vii) What is diploid parthenogenesis?

(viii) What are fraternal twins?

(ix) Write the plants in temperate deciduous Forests.

(x) Write a note on profundal zone.

(xi) Write a note on Tidal power.

(xii) What is reforestation?

Q3. Write short answers of any eight parts from the following.

2x8=16

(i) Define gene linkage. How does gene linkage affect variations among offsprings?

(ii) How are transgenic bacteria used to improve plant health? Give two examples.

(iii) What are different types of hormones on the basis of chemical nature?

(iv) Define food web. How do pathways of food web help to maintain stability of ecosystem?

(v) Enlist antibodies found in A, AB, B and O blood groups.

(vi) How plant growth is affected by ethene?

(vii) Differentiate between Phenotype and genotype with examples.

(viii) Write the structural components of limbic system

(ix) Define DNA finger printing. Write its significance.

(x) Define habitat and niche.

(xi) What is the significance of Transgenic Corn and Soybean?

(xii) Define mutualism. Give two examples.

Q4. Write short answers of any six parts from the following.

2x6=12

(i) What are Okazaki fragments? Give their lengths.

(ii) What is primitive streak? How is it formed?

(iii) Define Transcription and Anticodon.

(iv) What is meant by Nucleosome and gene?

(v) State Regeneration and dedifferentiation.

(vi) Define Interphase. Name its subphases.

(vii) What are vestigial organs? Give examples as well.

(viii) Characterize pachytene in Meiosis I.

(ix) Define genetic drift and hydrothermal vents.

Q. Section - II — 8x3=24

NOTE: Answer any three questions from the following.

Q5. (a) How does osmoregulation take place in terrestrial animals?

4

(b) What are different components of ecosystem?

4

Q6. (a) Discuss sliding filament model of Muscle contraction.

4

(b) Describe the process of transcription.

4

Q7. (a) Explain Feedback mechanism.

4

(b) Write a note on importance of forests.

4

Q8. (a) Describe the types of parthenogenesis in animals.

4

(b) What is dominance? Explain complete and incomplete dominance with examples.

4

Q9. (a) Describe in your own words the Growth Correlations in plants.

4

(b) Describe evidence of evolution from the Comparative Anatomy of animals.

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